

DG Lie Algebroid and Quotient Stack

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In this paper, I'll discuss foliation from point of view of derived algebraic geometry. Derived algebraic geometry proves the ∞ -equivalence of dg Lie algebroid and foliation. Now the quotient is defined by quotient of dg L_∞ groupoid generated from dg Lie algebroid, and it's calculated by dg L_∞ version of Chevalley-Eilenberg complex. I'm also wondering evaluation of the foliation using language of DT/GW invariants. First of all, the idea of foliation is classically defined differential geometrically on a smooth manifold:

Definition 0.1. Foliation

Foliation is a partition of a smooth manifold M into its submanifolds. One idea of generating foliation is to retrieve from distribution $D_x \subset TM$ for some $x \in M$, and this distribution classifies the points $x \in M$ to define foliation. Distribution D_x is defined by its vector field $X, Y \in \Gamma(D_x)$ satisfying $[X, Y] \in \Gamma(D_x)$, and in this case D_x is called integrable. This D_x corresponds to a submanifold L , since there's its restriction $D_x|_L = T_x L$ by the unique maximal submanifold $L \subset M$ s.t. $x \in L$. Note in fact $D_x|_{L_y} = T_y L$ for all $y \in L$. Accordingly, the resulting set $\{L_y\}$ for $y \in M$ is the set of submanifolds $L_y \subset M$ and disjoint union $\coprod L_y = M$.

To verbalize the notion of foliation is an equivalent notion with integrability by Frobenius theorem, where integrability is done by Lie algebra. If Lie algebra describes the global property (there's no "if" in math Toynbee says), the corresponding Lie group quotient of manifold $[M/G]$ is a stack. It doesn't work, however. I'll use Lie algebroid to describe every point of the submanifold $L \subset M$, whose Lie groupoid $\mathcal{G}$ quotient is a quotient stack, which describes foliation $[M/\mathcal{G}]$. I'll first of all terminology to define foliation derived algebraic geometrically.

Definition 0.2. Groupoid

A groupoid is a category in which every morphisms are invertible. Denote the objects are $Ob(\mathcal{C}) = G_0$, and the morphisms are $Mor(\mathcal{C}) = G_1$, the double arrow $G_1 \rightrightarrows G_0$. For an arrow $g : x \rightarrow y$, there are source and target maps $s, t : G_1 \rightarrow G_0$, together with composition $G_1 \times_{G_0} G_1 \rightarrow G_1$, identity arrow and inverses. A group becomes a Lie groupoid if G_0 and G_1 are smooth manifolds, and all the maps are smooth.

There are many important examples of Lie groupoid. The trivial things (logically equivalent to Lie group but somehow interesting generalization) are Lie group, which is a Lie groupoid with one object so $G \rightrightarrows *$. Another example which is more geometric is that Lie group action $G \times X \rightrightarrows X$ can be modeled by $G_1 = G \times X$ and $G_0 = X$. By the context of foliation, Lie groupoid $Hol(\mathcal{F})$ of the foliation $\mathcal{F}$ is holonomy groupoid. All the above examples are important. Now we consider quotient stack by Lie groupoid, in particular by Lie group:

Definition 0.3. Quotient Stack

For an algebraic variety X , quotient stack by Lie group G is $[X/G]$ is a functor $[X/G] : (Spaces)^{op} \rightarrow Groupoids$.

$$[X/G](S) = \{(P \rightarrow S, f : P \rightarrow X) | P \rightarrow S \text{ is a principal } G\text{-bundle, } f \text{ is } G\text{-equivariant}\}$$

In particular if $X = pt$, the quotient stack is $[pt/G] = BG$, and $BG(pt)$ is one-object groupoid whose automorphism group is $G = Hom(*, *)$.

Meanwhile, quotient stack by groupoid $[U/R]$ for a groupoid R is a stackification $[U/R] = St([U/_p R])$ of category fibered in groupoids $p : U \rightarrow R$. This works for all $c \in R$, there exists $x \in U$ s.t. $p(x) = c$, and for all $f : x \rightarrow y$, $p(f) = id_c$.

where I abbreviate what the stackification means in this introductory chapter for the moment. In any case,

the important quotient stack is holonomy groupoid quotient $[M/Hol(F)]$, which is a foliation. But how does Lie groupoid appears, where Lie algebra solves integrability condition in manifold? Lie algebra can be generalised to Lie algebroid by its globalization, where Lie groupoid comes. The definition of Lie algebroid is:

Definition 0.4. Lie Algebroid

For a smooth manifold M , a Lie algebroid over M consists of $(A \rightarrow M, [\cdot, \cdot]_A, \rho)$ where $A \rightarrow M$ is a vector bundle, $[\cdot, \cdot]_A : \Gamma(A) \times \Gamma(A) \rightarrow \Gamma(A)$ is a Lie bracket of sections, $\rho : A \rightarrow TM$ is an anchor map. These satisfy product rule $[X, fY]_A = f[X, Y]_A + \rho(X)(f)Y$, and the anchor preserves bracket $\rho([X, Y]_A) = [\rho(X), \rho(Y)]$.

The important remark is Lie algebroid doesn't always correspond to Lie groupoid unlike the Lie algebra/-group correspondence, since Lie groupoid always make Lie algebroid by infinitesimalization but not the other way around, since the integration of Lie algebroid contains obstruction. Here as many people already does it, I'll introduce technique of derived algebraic geometry to resolve this problem. There's several naïve questions:

Question 0.1

- How to extend Lie algebroid to Lie groupoid?
- What happens if the foliation is Lagrangian foliation?
- Is there any connection to Fukaya category?
- Is there any either deformation or quantization?

and some other stuffs. As somebody does it already, I'll introduce the technique of derived algebraic geometry to develop this foliation problem.

Definition 0.5

1 Prelim

Definition 1.1

References

- [1] D. Calaque, C. A. Rossi, and M. van den Bergh. "Hochschild (Co)homology for Lie Algebroids". In: *International Mathematics Research Notices* (Mar. 2010). ISSN: 1687-0247. DOI: 10.1093/imrn/rnq033. URL: <http://dx.doi.org/10.1093/imrn/rnq033>.
- [2] Alfonso Garmendia and Marco Zambon. *Quotients of singular foliations and Lie 2-group actions*. 2020. arXiv: 1904.08890 [math.DG]. URL: <https://arxiv.org/abs/1904.08890>.
- [3] Robin Hartshorne. *Algebraic Geometry*. Springer, 1977.
- [4] Isamu Iwanari. *Moduli theory associated to Hochschild pairs*. 2022. arXiv: 2104.02919 [math.AG]. URL: <https://arxiv.org/abs/2104.02919>.
- [5] Isamu and Iwanari. "Differential Calculus of Hochschild Pairs for Infinity-Categories". In: *Symmetry, Integrability and Geometry: Methods and Applications* (Oct. 2020). ISSN: 1815-0659. DOI: 10.3842/sigma.2020.097. URL: <http://dx.doi.org/10.3842/SIGMA.2020.097>.
- [6] Jacob Kryczka and Artan Sheshmani. *Tyurin Degenerations, Derived Lagrangians and Categorification of DT Invariants*. 2025. arXiv: 2510.20325 [math.AG]. URL: <https://arxiv.org/abs/2510.20325>.
- [7] Joan Milles. *The Koszul complex is the cotangent complex*. 2010. arXiv: 1004.0096 [math.AT]. URL: <https://arxiv.org/abs/1004.0096>.
- [8] Joost Nuiten. "Homotopical Algebra for Lie Algebroids". In: *Applied Categorical Structures* 27.5 (Feb. 2019), pp. 493–534. ISSN: 1572-9095. DOI: 10.1007/s10485-019-09563-z. URL: <http://dx.doi.org/10.1007/s10485-019-09563-z>.

- [9] Bertrand Toën and Gabriele Vezzosi. *Algebraic foliations and derived geometry: the Riemann-Hilbert correspondence*. 2020. arXiv: 2001.05450 [math.AG]. URL: <https://arxiv.org/abs/2001.05450>.
- [10] Bertrand Toën and Gabriele Vezzosi. “Homotopical Algebraic Geometry II: Geometric Stacks and Applications”. In: *Memoirs of the American Mathematical Society* 193.902 (2008).